



Contents lists available at ScienceDirect

American Journal of Emergency Medicine

journal homepage: www.elsevier.com/locate/ajemThe
American Journal of
Emergency Medicine

Original Contribution

Prognostic importance of neutrophil-lymphocyte ratio in critically ill patients: short- and long-term outcomes

Nazire Belgin Akilli, MD ^{a,*}, Mehmet Yortanlı, MD ^a, Hüseyin Mutlu, MD ^a, Yahya Kemal Günaydın, MD ^a, Ramazan Koylu, MD ^a, Hatice Seyma Akca, MD ^a, Emine Akinci, MD ^b, Zerrin Defne Dundar, MD ^c, Basar Cander, MD ^c

^a Department of Emergency Medicine, Konya Education and Research Hospital, Konya, Turkey

^b Department of Emergency Medicine, Keciören Education and Research Hospital, Ankara, Turkey

^c Department of Emergency Medicine, Necmettin Erbakan University, Meram Faculty of Medicine, Konya, Turkey

ARTICLE INFO

Article history:

Received 2 May 2014

Received in revised form 31 August 2014

Accepted 1 September 2014

Available online xxxx

ABSTRACT

Study objective: The number of critically ill patients admitted to the emergency department increases daily. To decrease mortality, interventions and treatments should be conducted in a timely manner. It has been found that the neutrophil-lymphocyte ratio (NLR) is related to mortality in some disease groups, such as acute coronary syndrome and pulmonary emboli. The effect of the NLR on mortality is unknown in critically ill patients who are admitted to the emergency department. Our aim in this study is to evaluate the effect of the NLR on mortality in critically ill patients.

Methods: This study was planned as a prospective, observational cohort study. Patients who were admitted to the emergency department because they were critically ill and required the intensive care unit were included in the study. Demographic characteristics, Acute Physiology and Chronic Health Evaluation II (APACHE II), Sepsis-related Organ Failure Assessment, Glasgow Coma Score, and NLR values were recorded upon emergency department admission. The patients were followed up for sepsis, ventilator-associated pneumonia, multiorgan failure, in-hospital mortality, and 6-month mortality.

Results: The median (interquartile range) age of the 373 patients was 74 (190) years, and 54.4% were men. Neutrophil-lymphocyte ratio values were divided into quartiles, as follows: less than 3.48, 3.48 to 6.73, 6.74–13.6, and more than 13.6. There was no difference among these 4 groups regarding demographic characteristics, APACHE II score, Sepsis-related Organ Failure Assessment score, Glasgow Coma Score, and length of hospital stay ($P > .05$). In the multivariable Cox regression model, in-hospital mortality and 6-month mortality NLR were hazard ratio (HR), 1.63 (1.110–2.415; $P = .01$) and HR, 1.58 (1.136–2.213; $P = .007$), respectively, and APACHE II scores were detected as independent indicators.

Conclusion: The NLR is a simple, cheap, rapidly available, and independent indicator of short- and long-term mortalities. We suggest that the NLR can provide direction to emergency department physicians for interventions, particularly within a few hours after admission, in the critically ill patient group.

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1. Introduction

Although advancements in science and technology have increased the average life expectancy, it has also created a patient group with high comorbidity within the advanced age group. Therefore, the number of critically ill patients in the emergency department is increasing daily, and these patients' medical conditions are more complicated, and their emergency department services required gets longer. In a study that compared the length of stay of patients in the emergency department during the years 1988 and 1997, the length of stay increased by 152% [1]. Early diagnosis and treatment of these patients in a timely

manner are of vital importance. The best example of this is early goal-directed therapy for sepsis [2]. According to previous studies, impaired physiologic parameters can be reversed with interventions in the emergency department, and these several hours are as equally important as the first 3 days in the intensive care unit (ICU). This state will be called the “golden hour” and “silver day [3–5].”

White blood cell (WBC) count is a parameter that is included in many scoring systems that are frequently used in daily practice for the diagnosis and follow-up of diseases. According to recent studies, important changes occur in WBC subtypes under stress. Galus and Stern [6] reported that extreme lymphocytopenia may occur during infectious emergencies, such as toxic shock syndrome. Jilma et al [7] detected an increase in the number of neutrophils and a decrease in monocytes after inflammation. Along with these changes, Zahorec [8] identified a new parameter known as the neutrophil-lymphocyte stress factor.

* Corresponding author. Konya Eğitim ve Araştırma Hastanesi Acil Servis Bölümü 42080–Konya, Turkey. Tel.: +90 332 2236409; fax: +90 332 2237941.

E-mail address: drbelginakilli@hotmail.com (N.B. Akilli).

According to this parameter, when the ratio of peripheral neutrophils to lymphocytes (NLR) is compared with the WBC count, a better prognostic indicator is achieved. Furthermore, studies based on acute coronary syndrome, congestive heart failure, and pulmonary emboli have demonstrated that the NLR is related to mortality [9–12]. The relationship of the NLR to mortality is unknown in critically ill patients.

In this study, we aimed to evaluate the effect of the NLR on in-hospital mortality and 6-month mortality in critically ill patients who were admitted to the emergency department.

2. Materials and methods

2.1. Study design and setting

This prospective, observational cohort study was conducted in a regional academic hospital emergency department that provides tertiary care in Konya. This emergency department provides care to approximately 300 000 patients annually, and 2.6% of the patients are admitted to the ICU. The study was conducted between January 1, 2013, and August 10, 2013. Approval from the local ethics committee was obtained for the study and is compatible with the Helsinki Declaration.

2.2. Selection of participants

Critically ill adult patients were included in the study and were defined as patients accepted by the emergency department who had 2 of the 4 systemic inflammatory response syndrome criteria: heart rate more than 90 beats per minute, respiratory rate more than 20 breaths per minute, temperature more than 38°C or less than 36°C, WBC count more than 12 000 cells/mm³ and less than 4000 cells/mm³ or more than 10% band cells as well as systolic blood pressure below 90 mm Hg or 2 of the systemic inflammatory response syndrome criteria and a lactate level above 4 mmol/L [13]. Exclusion criteria were age younger than 18 years, pregnancy, known hematologic disease, previous chemotherapy treatment (within the last month), blood transfusion (within the last 2 weeks), chronic hepatic disease, trauma, or poisoning.

2.3. Outcome measures

Demographic data, comorbidities, and diagnoses of all patients were recorded. Vital findings upon emergency department admission, laboratory findings, and neutrophil and lymphocyte values from whole blood counts were taken. Neutrophil-lymphocyte ratio, Acute Physiology and Chronic Health Evaluation II (APACHE II) [14], Sepsis-related Organ Failure Assessment (SOFA) [15], and Glasgow Coma Score (GCS) scores were calculated. After the patients were accepted to the ICU from the emergency department, they were evaluated for the development of sepsis, ventilator-associated pneumonia, and multiorgan failure (MOF). Duration of hospital stay was recorded. In-hospital and 6-month mortalities of the patients were determined from medical records. In the cases where data were not available from the records, data were obtained via phone call. The primary end point was in-hospital mortality and 6-month mortality; secondary end points were sepsis, ventilator-associated pneumonia, and MOF, respectively.

2.4. Blood analysis

Complete blood counts and differentials were studied in the peripheral blood samples. Blood samples were taken in calcium-EDTA tubes. Complete blood counts were performed with an autoanalyzer (SIEMENS ADVIA 120 hematology analyzer; SIEMENS, Eschborn, Germany). The NLR was calculated after the whole blood count NLR.

2.5. Data analysis

Statistical analysis was performed using SPSS version 15.0 for Windows (SPSS, Chicago, IL). Both visual (histogram and probability graphs) and analytical (Kolmogorov-Smirnov and Shapiro-Wilk tests) methods were used to determine if the data were normally distributed. Descriptive variables are expressed as the mean $\pm$ SD for data that are normally distributed and as the median and interquartile range (IQR) for variables that are not normally distributed. The χ^2 or Fisher exact test was used to compare categorical values. Clinical and laboratory characteristics, according to the NLR quartile, were evaluated via one-way analysis of variance for normally distributed variables, whereas the Kruskal-Wallis test was used for variables without a normal distribution. When necessary, the Mann-Whitney *U* test with the Bonferroni correction was used to compare variables. Univariate and multivariate Cox regression models were used to evaluate the independent relationships of in-hospital and 6-month mortalities with NLR values. Age, sex, congestive heart failure, coronary artery disease, diabetes mellitus, malignancy, corticosteroid treatment, creatinine, hematocrit, WBC, APACHE II score, and SOFA scores were included in this model. This model was considered to be 11.9 cutoff value of NLR. The effect of the NLR on the survival of critically ill patients was investigated using the log-rank test. Kaplan-Meier survival estimates were calculated. The utility of the NLR in long-term mortality in critically ill patients was evaluated via receiver operating characteristic curves, and the cutoff value was determined using Youden's index. A $P < .05$ was considered statistically significant. Study power was calculated as 92% using a free Web site (http://www.statisticalsolutions.net/pss_calc.php).

3. Results

A total of 373 patients were included in the study. Of the 373 patients, 203 were men (54.4%) and 170 were women (45.6%). The median (IQR) age was 74 (19) years. The mean APACHE II score was 20.7 ± 10.1 , the SOFA score was 4.9 ± 2.2 , and the median (IQR) NLR was 6.72 (10.3). Demographics and clinical characteristics of the patients are provided in Table 1.

In the comparison, according to NLR quartiles, there were no differences detected for age, sex, APACHE II, SOFA score, GCS, comorbid diseases, mechanical ventilation, and duration of hospital stay ($P > .05$). Because there was not any difference for ventilator-associated pneumonia time on the ventilator in terms of NLR quartiles, the development of MOF was at significantly higher levels in the fourth quartile than the other quartiles ($P = .04$) (Table 2).

Table 1
Demographics and laboratory findings of the study population

No. of patients	373
Age, y, median (IQR)	74 (19)
Female sex, n (%)	170 (45.6)
Diabetes mellitus, n (%)	153 (41.0)
Coronary artery disease, n (%)	78 (20.9)
Congestive heart failure, n (%)	83 (22.3)
Chronic renal failure, n (%)	40 (10.7)
Chronic obstructive pulmonary disease, n (%)	130 (34.9)
Malignancy, n (%)	21 (5.6)
Sepsis, n (%)	35 (9.4)
Corticosteroid treatment, n (%)	34 (9.1)
APACHE II score, mean $\pm$ SD	20.7 ± 10.1
SOFA score, mean $\pm$ SD	4.9 ± 2.2
GCS, median (IQR)	12 (8)
Creatinine, milligrams per deciliter, median (IQR)	1.17 (1.09)
Hematocrit, percent, mean $\pm$ SD	39.5 ± 8.8
WBC count, thousands per microliter, median (IQR)	12.8 (7.02)
Neutrophil, thousands per microliter, median (IQR)	9.0 (6.1)
Lymphocyte, thousands per microliter, median (IQR)	1.3 (1.4)
NLR, median (IQR)	6.72 (10.3)
Length of stay in-hospital, d, median (IQR)	6.0 (9.1)

Table 2

Patients characteristics according to quartile of NLR

	<3.48	3.48–6.73	6.74–13.6	>13.6	P
Age, y, median (IQR)	70 (25)	73.5 (29)	74 (19)	77 (15)	.06
Female sex, n (%)	48 (51.1)	44 (47.3)	38 (40.4)	40 (43.5)	.49
Diabetes mellitus, n (%)	39 (41.5)	43 (46.2)	28 (29.8)	43 (46.7)	.66
Coronary artery disease, n (%)	25 (26.6)	15 (16.1)	14 (14.9)	24 (26.1)	.08
Congestive heart failure, n (%)	26 (27.7)	18 (19.4)	16 (17)	23 (25)	.27
Chronic renal failure, n (%)	10 (10.6)	5 (5.4)	14 (14.9)	11 (12)	.20
Malignancy, n (%)	2 (2.1)	6 (6.5)	6 (6.4)	7 (7.6)	.30
Corticosteroid treatment, n (%)	8 (8.5)	9 (9.6)	7 (7.4)	10 (10.8)	.59
APACHE II score, mean $\pm$ SD	21.3 $\pm$ 10.1	19.8 $\pm$ 9.6	20.7 $\pm$ 9.9	20.9 $\pm$ 8.9	.75
SOFA score, mean $\pm$ SD	4.4 $\pm$ 2.2	5.21 $\pm$ 2.4	4.9 $\pm$ 2.2	4.8 $\pm$ 2.0	.27
GCS, median (IQR)	13 (12)	12 (8)	10 (11)	13 (5)	.38
Development of sepsis, n (%)	22 (23.7)	16 (17.2)	17 (18.1)	29 (31.5)	.07
Creatinine, milligrams per deciliter, median (IQR)	1.09 (1.01)	1.06 (0.98)	1.17 (1.11)	1.52 (1.32)	.02
Hematocrit, percent, mean $\pm$ SD	39.5 $\pm$ 9.3	39.6 $\pm$ 9.4	39.5 $\pm$ 8.5	39.8 $\pm$ 7.8	.99
WBC count, thousands per microliter, median (IQR)	10.2 (8.02)	10.4 (6.15)	11.6 (6.41)	13.6 (7.78)	<.001
Application of MV, n (%)	50 (53.2)	38 (40.9)	44 (46.8)	53 (57.6)	.11
Duration of MV, d, median (IQR)	1 (6)	0.5 (4)	1 (2)	1 (5)	.62
Ventilator-associated pneumonia, n (%)	21 (22.6)	14 (15.1)	14 (14.9)	19 (20.7)	.41
MOF, n (%)	62 (65)	50 (53.7)	51 (54.2)	65 (70.7)	.04
Length of stay in-hospital, d, median (IQR)	6 (9)	6 (10)	5 (8)	7 (9)	.52

Abbreviation: MV, mechanical ventilator.

There was not any correlation detected between the NLR and age, APACHE II, SOFA, duration of hospital stay, or duration of mechanical ventilation ($P > .05$).

Performance characteristics (area under the curve, sensitivity, specificity, predictive value, and likelihood ratios) of the NLR for predicting mortality were calculated (Table 3). The cutoff value for mortality obtained using the ROC curve was 11.9 (sensitivity, 37.0%; specificity, 81.8%; area under the curve, 0.61; 95% confidence interval [CI], 0.55–0.65; $P = .01$) (Fig. 1).

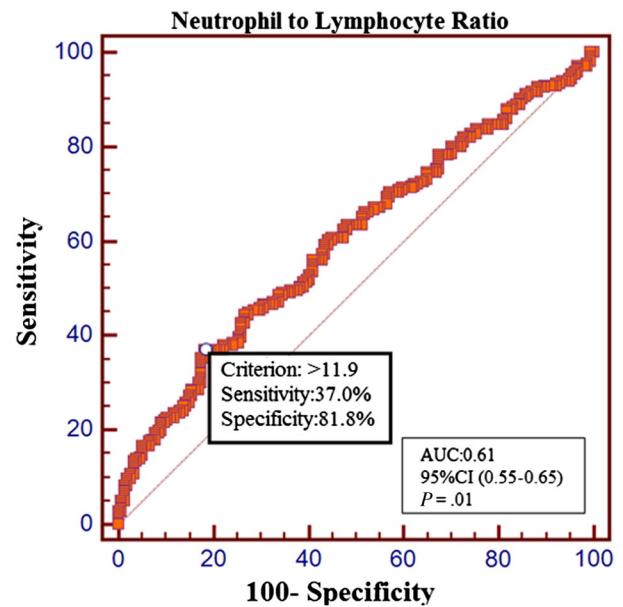
Sepsis development, mechanical ventilation, and ventilator-associated pneumonia were analyzed according to the cutoff value. When the NLR is less than 11.9, sepsis developed in 19.5% (50) of the patients, whereas when the NLR is greater than or equal to 11.9, sepsis development significantly increased to 29.6% (34) ($P = .03$). However, there was not a significant difference detected for mechanical ventilation and ventilator-associated pneumonia development ($P > .05$).

During follow-up, 164 mortalities occurred in the hospital. When the dead and alive patients were compared, NLR values were found to be

Table 3

Performance parameters of NLR as a predictor of mortality

Performance parameters	95% confidence interval
Sensitivity	37.04 (30.6–43.9)
Specificity	81.8 (74.8–87.6)
Positive predictive value	74.1 (64.7–82.1)
Negative predictive value	48.1 (41.9–54.3)
Positive likelihood ratio	2.04 (1.4–3.0)
Negative likelihood ratio	0.77 (0.7–0.9)
Receiver operating characteristic curve area	0.61 (0.55–0.65)

**Fig. 1.** Receiver operating characteristic curve of NLR to predict mortality.

significantly different ($P = .04$). When the in-hospital mortalities were compared according to the NLR quartiles, the fourth quartile (52 [31.7%]) had significantly higher mortality rates than the first, second, and third quartiles (38 [23.2%], 39 [23.8%], and 35 [21.3%], respectively; $P < .041$). Survival rates for in-hospital mortality, calculated with the Kaplan-Meier curve, were almost significantly lower in the fourth quartile (log-rank, 7.82; $P = .05$), and survival rates were significantly lower for 6-month mortality (log-rank, 10.45; $P = .01$) (Fig. 2). In the multivariable Cox regression model, for in-hospital mortality and 6-month mortality, NLR was (Hazard ratio (HR), 1.637 [1.110–2.415]; $P = .01$ and HR, 1.585 [1.136–2.213]; $P < .007$, respectively). In this model, accepted cutoff value of NLR was 11.9. Acute Physiology and Chronic Health Evaluation II scores were detected as independent indicators (Table 4). In the Kaplan-Meier curve of patients with NLR more than 11.9, in-hospital mortality and 6-month mortality had significantly lower survival rates. (log-rank, 5.61; $P = .002$ and log-rank, 9.12; $P = .018$, respectively) (Figs. 3 and 4).

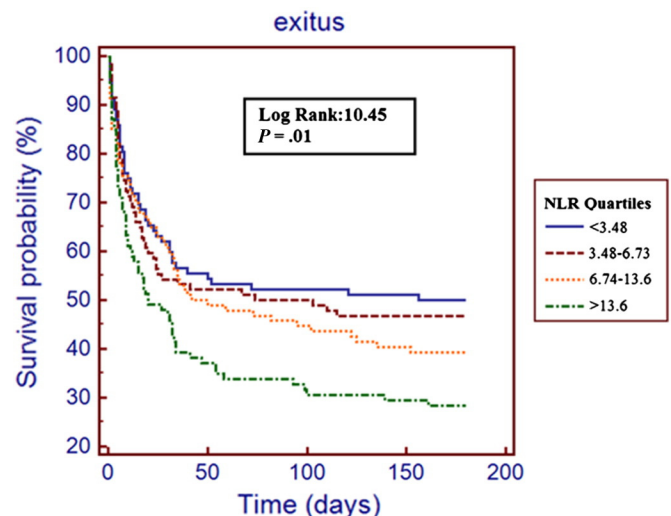
**Fig. 2.** Comparison of Kaplan-Meier survival curves of NLR quartiles.

Table 4
The independent predictors of mortality by multivariate logistic regression analysis

Predictor	Hazard ratio	95% CI	P
In-hospital mortality			
NLR	1.637	1.110–2.415	.01
APACHE II	1.084	1.061–1.108	<.001
180-Day mortality			
NLR	1.585	1.136–2.213	.007
APACHE II	1.082	1.064–1.100	<.001

4. Discussion

This study was a prospective cohort study evaluating the prognostic value of the NLR in the critically ill patient group. According to the data we obtained from this study, high NLR levels in critically ill patients are an independent indicator of in-hospital and 6-month mortality. Neutrophil-lymphocyte ratio levels more than 11.9 increase in-hospital mortality by 64% and increase 6-month mortality by 59%. Furthermore, the risks of MOF and sepsis development are higher with higher NLR values. To our knowledge, this study is the first prospective study that presents the prognostic value of the NLR in critically ill patients in the emergency department.

The diagnosis and resuscitation of critical patients are the reason for emergency service. Each day, the number of patients admitted to emergency departments with complex geriatric syndromes or decompensation of chronic diseases rises [16]. In particular, 19% of emergency department admissions consisted of elderly patients in 2000, and this rate is estimated to reach 34% by 2050 [17]. Most of our study group was elderly. Advancements in science and technology have prolonged life expectancy and increased the number of comorbid patients. In the study by Vinton et al [18], patients were grouped according to the number of emergency department admissions they had in 1 year, and they found that patients who were admitted to the emergency department most frequently (≥ 10 visits/year) were the patients with chronic disease and low socioeconomic status. Critical care services provided to this complex, high-volume patient group in the emergency department, and ICUs compose most health expenditures. In spite of scoring systems were improved, which are quality of intervention to patients can be evaluated and be able to predict prognosis, cheap and fast biological markers are very less. Because interventions performed on critical patients within the first several hours of admission play a very important

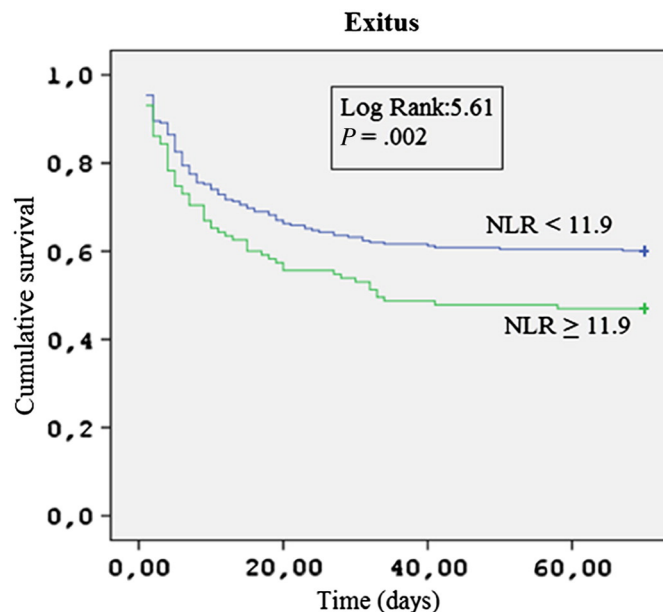


Fig. 4. Comparison of Kaplan-Meier survival curves of NLR cutoff for in-hospital mortality.

role in reducing mortality and the first several hours in the emergency department are equal to 72 hours in the ICU, they are known as the “golden hour and silver day” proposal [3–5]. In this case, the importance of markers for prognosis increases.

White blood cell number is a frequently used parameter that is included in many scoring systems and is frequently used for the diagnosis and follow-up of disease in daily practice. White blood cells play a main role in the systemic inflammatory response [8]. Jilma et al [7] researched the changes in WBC types after inflammation and detected a 300% increase in circulating neutrophils, 96% decrease in monocytes, and 85% decrease in lymphocytes 4 to 6 hours after inflammation. Dionigi et al [19] and Ayala et al [20] summarized the possible reasons of lymphocytopenia as induction of catecholamine, prolactin, and cortisol levels that increase with stress, margination of lymphocytes to the reticuloendothelial system, and apoptosis. Menges et al [21] explain the possible reasons for neutrophilia as demargination of neutrophils from the endothelium, delay of neutrophil apoptosis, and effect of growth factors on stem cells. Zahorec [8] reported a study conducted with 90 intensive care patients who developed neutrophilia and lymphocytopenia; however, the NLR was more valuable, and the authors state that this ratio is a reliable parameter for reflecting the intensity of stress, evaluating systemic inflammatory response, and monitoring.

Recent studies regarding the NLR mostly suggest that the NLR is a prognostic factor for cardiovascular diseases. Uthamalingam et al [11] found that in acute decompensated heart failure, NLR at admission was related to 30-day mortality, and they recommended the use of NLR for risk stratification. Thus, it has been demonstrated in studies that NLR is a prognostic factor for acute coronary syndromes [9,10]. Kayrak et al [12] found that NLR is an independent indicator of short-term mortality in pulmonary emboli at the NLR greater than or equal to 9.2 cutoff value (HR, 3.52 [1.36–9.10] 95% CI; $P = .01$). According to our literature search, we did not find a similar study for the critically ill patient group.

Our results demonstrate that in the critically ill patient group, NLR (HR, 1.637 [1.110–2.415]; $P < .05$ and HR, 1.585 [1.136–2.213]; $P < .05$) and APACHE II (HR, 1.084 [1.061–1.108]; $P < .001$ and HR, 1.082 [1.064–1.100]; $P < .001$) can be independent indicators of short- and long-term mortality. The NLR is a simple, cheap, and rapidly available parameter. In our opinion, the NLR can be used to determine the prognosis of critical patients in the emergency department. Because it is rapidly obtainable, the NLR can provide knowledge to perform the necessary interventions within the very important golden hours. For

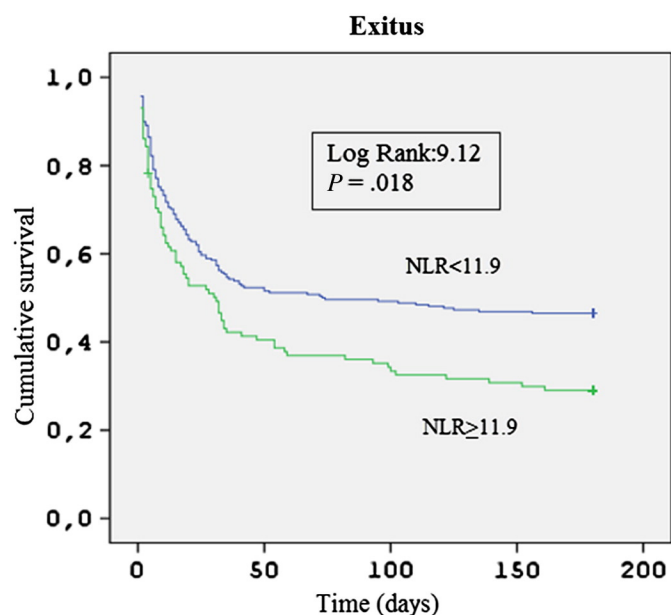


Fig. 3. Comparison of Kaplan-Meier survival curves of NLR cutoff for 6-month mortality.

us, this will be used as a biological marker to determine the prognosis on critical patients in emergency department.

Zahorec [8] demonstrated that the clinical improvement of major surgery, sepsis, and injury parallels an increase in the number of lymphocytes and a decrease in the number of neutrophils, and persistent neutrophilia and lymphopenia may be related to development of MOF. Our results also demonstrated that the development of MOF in the fourth quartile was significantly higher ($P = .03$). For NLR less than 11.9, sepsis developed in 19.5% (50) of the patients, whereas for NLR greater than or equal to 11.9, sepsis development significantly increased to 29.6% (34) ($P = .03$). However, there was not a significant difference detected for mechanical ventilation and ventilator-associated pneumonia development ($P > .05$). The NLR may not only shed light on short- and long-term mortality but can also enlighten the physician to the development of MOF and sepsis.

Zahorec [8] performed the aforementioned study with oncology ICU patients who had sepsis or septic shock. Only 10% of our patient group was critical due to infectious causes. Therefore, the NLR cannot only be used in infectious cases as a prognostic factor but can also be used in noninfectious critical patient groups.

5. Limitations

Our study had several limitations. The first limitation was that the study was a single-center study. Our hospital is a regional hospital, where medical intensive care patients are frequently accepted. For example, chronic obstructive pulmonary disease patients are very common in our region and constitute a major portion of the patients admitted to our emergency department, particularly in the winter months. For this reason, multicenter studies, including trauma centers, are needed for further studies.

The second limitation is that repetitive measurements of NLR were not performed. The effects of repetitive measurements on determining the effectiveness of treatment and the effect of the change on mortality are unknown.

6. Conclusion

This study investigated the prognostic value of the NLR in the critical patient group. The NLR, which is a simple measure, does not require additional costs, is rapidly accessible, and is an independent indicator of short- and long-term mortality. We suggest that the NLR can direct emergency department physicians toward interventions within several hours of admission in the critical patient group.

References

- [1] Meggs W, Czapliński T, Benson N. Trends in emergency department utilization, 1988–1997. *Acad Emerg Med* 1999;6:1030–5.
- [2] Rivers E, Nguyen B, Havstad S, Ressler J, Muzzin A, Knoblich B, et al. Early goal directed therapy in the treatment of severe sepsis and septic shock. *NEJM* 2001;345:1368–77.
- [3] Lundberg JS, Perl TM, Wiblin T, Costigan MD, Dawson J, Nettleman MD, et al. Septic shock: an analysis of outcomes for patients with onset on hospital wards versus intensive care units. *Crit Care Med* 1998;26:1020–4.
- [4] Garrard C, Young D. Suboptimal care of patients before admission to intensive care is caused by a failure to appreciate or apply the ABCs of life support (editorial). *BMJ* 1998;316:1841–2.
- [5] Blow O, Magliore L, Claridge JA, Butler K, Young JS. The golden hour and the silver day: detection and correction of occult hypoperfusion within 24 hours improves outcome from major trauma. *J Trauma* 1999;47:964–9.
- [6] Galus MA, Stern J. Extreme lymphocytopenia associated with toxic shock syndrome. *J Intern Med* 1998;244:351–4.
- [7] Iljina B, Blann A, Pernsterstorfer T, Stohlawetz P, Eichler HG, Vondrovec B, et al. Regulation of adhesion molecules during human endotoxemia. *Am J Respir Crit Care Med* 1999;159:857–63.
- [8] Zahorec. Ratio neutrophil to lymphocyte counts-rapid and simple parameter of systemic inflammation and stress in critically ill. *Bratisl Lek Listy* 2001;102:5–14.
- [9] Tamhane UU, Aneja S, Montgomery D, Rogers EK, Eagle KA, Gurm HS. Association between admission neutrophil to lymphocyte ratio and outcomes in patients with acute coronary syndrome. *Am J Cardiol* 2008;102:653–7.
- [10] Azab B, Zaher M, Weiserbs K, Estelle T. Usefulness of neutrophil to lymphocyte ratio in predicting short and long term mortality after non-ST elevation myocardial infarction. *Am J Cardiol* 2010;106:470–6.
- [11] Uthamalingam S, Patvardhan EA, Subramanian S, Ahmed W, Martin W, Daley M, et al. Utility of the neutrophil to lymphocyte ratio in predicting long-term outcomes in acute decompensated heart failure. *Am J Cardiol* 2011;107:433–8.
- [12] Kayrak M, Erdoğan H, Solak Y, Akilli H, Gül EE, Yıldırım O, et al. Prognostic value of neutrophil to lymphocyte ratio in patients with acute pulmonary embolism: a retrospective study. *Heart Lung Circ* 2014;23:56–62.
- [13] Nguyen HB, Rivers EP, Havstad S, Knoblich B, Ressler JA, Muzzin AM, et al. Critical care in the emergency department: a physiologic assessment and outcome evaluation. *Acad Emerg Med* 2000;7:1354–61.
- [14] Knaus Knaus WA, Draper EA, Wagner DP, Zimmerman JE. APACHE II: a severity of disease classification system. *Crit Care Med* 1985;13:818–29.
- [15] Vincent JL, Moreno R, Takala J, Willatts S, De Mendonça A, Bruining H, et al. The SOFA (Sepsis-related Organ Failure Assessment) score to describe organ dysfunction/failure. On behalf of the Working Group on Sepsis-Related Problems of the European Society of Intensive Care Medicine. *Intensive Care Med* 1996;22:707–10.
- [16] Arendts G, Lowthian J. Demography is destiny: an agenda for geriatric emergency medicine in Australasia. *Emerg Med Australas* 2013;25:271–8.
- [17] Martin-Khan M, Burkett E, Schnitker L, Jones RN, Gray LC. Methodology for developing quality indicators for the care of older people in the emergency department. *BMC Emerg Med* 2013;13:23–33.
- [18] Vinton DT, Capp R, Rooks SP, Abbott JT, Ginde AA. Frequent users of US emergency departments: characteristics and opportunities for intervention. *Emerg Med J* 2014;1–7.
- [19] Dionigi R, Dominioni L, Benevento A, Giudice G, Cuffari S, Bordone N, et al. Effects of surgical trauma of laparoscopic vs. open cholecystectomy. *Hepatogastroenterology* 1994;41:471–6.
- [20] Ayala A, Herdon CD, Lehman DL, Ayala CA, Chaudry IH. Differential induction of apoptosis in lymphoid tissues during sepsis: variation in onset, frequency, and the nature of the mediators. *Blood* 1996;15:4261–75.
- [21] Menges T, Engel J, Welters I, Wagner RM, Little S, Ruwoldt R, et al. Changes in blood lymphocyte populations after multiple trauma: association with posttraumatic complications. *Crit Care Med* 1999;27:733–40.